Technology & AI Architecture Overview

Fibre Assurance & Network Operations Platform

The platform combines a deterministic spatial and optical physics core (Python, FastAPI, PostgreSQL/PostGIS, NetworkX, binary OTDR codecs) with a grounded, multi-layer AI stack—delivering sub-second alarm triage, semantic runbook retrieval, and cited incident synthesis in cloud or air-gapped carrier environments.

End-to-End Pipeline

1. INGEST
Network & IT Feeds
Polls GIS, DWDM, RAN & BSS APIs; listens to live Syslog, SNMP & binary .sor OTDR files.

2. CORRELATE
Graph & Physics Core
Clusters multi-vendor storms, subtracts manhole slack on PostGIS & scores fault hypotheses.

3. RETRIEVE & REASON
Vector RAG & LLMs
Matches SOPs via pgvector and drafts cited briefs, RFOs & RFP checks from verified facts.

4. EXECUTE
Human-Gated Actions
Stages Jira/ServiceNow dispatches and disjoint optical reroutes behind NOC engineer approval.

AI & Machine Intelligence Stack

1. Grounded LLMs (NOC Copilot & RFO Synthesis)

Claude Sonnet 4 · GPT-4o · On-Prem Llama 3.3 / Qwen 2.5 (vLLM)

Generates cited NOC incident briefs (`[Alarm ID]`, `[Manhole ID]`), repair MOPs, and EN/Arabic customer RFOs strictly from verified JSON inputs, with a deterministic offline fallback.

2. Vector Embeddings & Runbook RAG

PostgreSQL pgvector · text-embedding-3 / BGE / 64-D Telecom Lexicon

Embeds live alarm signatures and runs cosine-similarity retrieval (⟺) against approved carrier SOPs and historical resolved incident tickets to surface proven remediation steps.

3. Causal Graph & Probabilistic Classifiers

NetworkX Connected-Subgraph Clustering + Confidence Scoring

Partitions 10,000-alarm storms across 6 network layers into isolated incidents and scores root-cause probability (fibre cut vs. GPON dying-gasp power loss, battery drain, or rain fade).

4. Optical & Vibration Signal Anomaly Detection

NumPy Waveform & Slope Analytics (.sor, Pre-FEC BER, SOP rad/s)

Detects mechanical excavator vibration along ducts, non-reflective splice pinches, and Pre-FEC BER cliff slopes 15–40 minutes before complete span Loss-of-Light occurs.

Core Platform Stack

LAYER	TECHNOLOGIES	ROLE IN PRODUCTION
Web & GIS UI	`React 18`, `TypeScript 5`, `MapLibre WebGL`	60 fps street duct/chamber maps, OTDR reflectometry curves, 5-layer blast radius, and approval gates.
Backend & Codecs	`Python 3.11+`, `FastAPI`, `Pydantic v2`, `SR-4731`	Async APIs, pluggable fault-classifier registry, Dijkstra diverse routing, and binary `.sor` + CRC-16 parsing.
Spatial & Vector DB	`PostgreSQL 16`, `PostGIS 3.4`, `pgvector`	Normalized 6-layer inventory, `WGS84` slack-adjusted trench interpolation, and vector runbook index.
Event Streaming	`Redpanda` (Kafka API), `asyncio` UDP	High-throughput alarm ingestion (`:19092`) plus native `RFC 5424` Syslog (`:15514`) & SNMPv2c (`:15162`).

Supported Telecom & IT Interfaces

NETWORK DOMAIN	REPRESENTATIVE SYSTEMS	PROTOCOLS & FORMATS
Optical / DWDM	Ciena MCP, Nokia NSP	`RESTCONF`, `OpenConfig gNMI`, `TMF642`
Fibre Testing (OTDR)	VIAVI ONMSi, EXFO Nova Fiber	Binary `Telcordia SR-4731 Rev 2 (.sor)`
Outside-Plant GIS	Lepton SmartInventory, ESRI, QGIS	`OGC WFS 2.0 GeoJSON` (ducts, chambers, slack)
5G RAN & Access	Ericsson ENM, GPON OLTs / GenieACS	`3GPP TS 32.435 XML`, `ITU-T G.984` Dying-Gasp
IP Core & Subsea	Cisco, Juniper, Subsea CLS PFE	`UDP Syslog`, `SNMPv2c`, PFE & C-OTDR telemetry
ITSM, DCIM & BSS	ServiceNow, Jira SM, NetBox, Comviva	`REST v3 / Table API`, `TMF629` & `TMF638` SLAs

AI Safety & Engineering Guardrails

1. **Zero Hallucinated Physics** — GPS dig coordinates, manhole slack subtraction, and disjoint backup wavelengths are computed strictly by PostGIS, binary OTDR math, and NetworkX—never estimated by generative AI.
2. **Evidence-Bound Prompts** — LLMs receive structured JSON outputs from the correlation and PostGIS engines and must cite exact `[Alarm ID]`, `[Runbook ID]`, and `[Manhole ID]` references.
3. **Sovereign & Air-Gapped Ready** — Supports cloud APIs (`Claude Sonnet 4`, `GPT-4o`), self-hosted open-weight models (`Llama 3.3` / `Qwen 2.5` via `vLLM`), and a zero-key deterministic synthesizer.
4. **Topology-Agnostic Code** — Application logic contains zero hardcoded cable, duct, or router IDs (enforced on every commit in GitHub Actions CI). Any operator's GIS dataset loads via configuration.
5. **Human-in-the-Loop Execution** — Southbound ingestion is strictly read-only. Optical wavelength switches and field work orders are staged as drafts and execute only after explicit NOC approval.

Runtime & Lab Emulator: Packaged with `Docker Compose` (PostgreSQL 16 + PostGIS + pgvector on `:5433`, Redpanda Kafka on `:19092`), FastAPI backend (`:8002`), standalone Multi-OEM lab emulator (`:8001`), and React web UI (`:5173`). Deploys unchanged to standard Kubernetes (`EKS / AKS / OpenShift`) or on-prem Linux VMs with GPU or CPU-only inference.